

of the bid-ask bounce. At this time, an investor enters the market and submits a sizable buy order that immediately pushes the price to \$30.25. The premium above current market price serves to attract additional sellers to complete the entire position. Price reversion immediately follows this transaction; but the price reverts to \$30.05 not the original price of \$30.00. Market participants inferred this trade as information based, due likely to the stock being undervalued. As a result, participants looking to sell additional shares were no longer willing to sell shares at \$30.00 but would be willing to offer the shares at \$30.05—what they perceive to be the new fair value. The investor incurred \$0.25 of total market impact with \$0.20 temporary impact (liquidity needs) and \$0.05 permanent impact (information content).

Figure 4.1b illustrates the same concept but for a sell order. The stock begins trading at \$30.00 and fluctuates between \$29.99 and \$30.01 due to the bid-ask bounce. An investor with a large order enters the market and immediately pushes the price down to \$29.75. The investor needed to discount the price to attract additional buyers. After the transaction, we once again observe price reversion but the price only returns to \$29.95, not \$30.00. Market participants believed the price was overvalued causing them to re-establish a fair value price. Total cost to the investor is \$0.25 with \$0.20 being temporary (liquidity needs) and \$0.05 being permanent (information content).

Illustration 2—Supply-Demand Equilibrium

We present a second illustration of the market impact concept through traditional economic supply-demand curves (Figure 4.2a–d). We use these curves to show the effect of a buy order on the stock price. Figure 4.2a depicts a system currently in equilibrium with q^* shares transacting at price of p^* . Figure 4.2b shows the effect of a new buyer entering the market on the equilibrium of the system. Assume the new buyer wishes to transact an incremental Δq shares. This results in a shift in the demand curve from D to D' to reflect the increased demand $q_1 = q^* + \Delta q$. It appears that the new equilibrium price for q_1 shares is p_1 but this is incorrect. Immediately after the new buyer enters the market, the group of existing sellers is likely to believe the demand was driven because the market price was undervalued and they will raise their selling prices. This results in an upward shift in the supply curve from S to S' and causes the price to increase to p_2 from p_1 (Figure 4.2c) for q_1 shares. The impact from the incremental demand of Δq is $p_2 - p^*$. After the trade, participants re-evaluate the price due to the information content of the trade. Their belief is likely to be that the incremental demand was due to the price